

RiskPulse

Run the project, understand the engine, and prepare the submission.

This package implements the unified AI/NLP Risk Engine and **Module A: Tactical Index Rebalancing**. It includes a local dashboard, API, synthetic replay, tests, deck, and demo scripts.

1. Start on Windows

Extract All: Download the ZIP, right-click it, choose Extract All, and open the riskpulse folder.

Python: Install Python 3.11 or newer if it is missing. Enable Add Python to PATH during installation.

Launch: Double-click **start.bat**. The first run installs the base dependencies. Keep that terminal open.

Open: Type **http://127.0.0.1:8000** in Chrome or Edge.

Try it: Choose Synthetic replay and run the scenario. Open Index allocation to see the saved weights.

2. Enable the full FinBERT model

The base app can use a clearly labeled financial-lexicon fallback. For the full NLP demonstration, stop the server with Ctrl+C after the base environment exists. Double-click **scripts/enable_finbert.bat**, wait for installation/download, then restart start.bat.

The first model download needs internet and about 438 MB for the weight file, plus dependency space. After caching, CPU inference and synthetic replay work locally.

Before recording: open Policy settings and confirm the active backend says **finbert**. Then reset/replay to regenerate the synthetic signals with that model.

macOS or Linux

Open a terminal in the riskpulse folder and run **bash start.sh**. The README contains the exact optional CPU FinBERT installation commands. Node is unnecessary for the bundled dashboard.

What the project does

News text becomes structured signals, then a controlled mock allocation.

Financial news comes from BBC Business. Company announcements come from Apple Newsroom and Microsoft's corporate blog. Optional Yahoo and GDELT connectors are also included. The engine analyzes headlines and available summaries.

The required risk outputs

Field	Meaning	Actual synthetic example
Sentiment Score	Financial tone, -1 to +1	-0.9585 (negative)
Event Classification	Dominant event category	Credit Event
Impact Score	Rule-based severity, 1 to 10	10 / 10

Example: a **fictional** JPMorgan liquidity crisis and missed repayment. FinBERT produced 96.73% class confidence. The example is synthetic and does not claim the company experienced this event.

Processing steps

Fetch and clean: preserve provider/source/time, remove HTML, and skip duplicate normalized headlines.

Analyze: aliases map companies. FinBERT estimates sentiment. Phrase rules classify the event and explain severity.

Rebalance: fresh, confident company signals tilt an equal-weight ten-stock index. Weight limits and turnover constrain changes.

Save and inspect: SQLite stores signals and changed allocations. The dashboard and JSON/CSV API expose the same records.

Live feeds and synthetic replay

Live mode contains provider and analyst records. Replay contains clearly labeled fictional scenarios and its own portfolio. A failed live fetch remains an error and never receives synthetic replacement data.

Sentiment describes tone. Impact is an uncalibrated severity proxy. Neither field guarantees a stock return. The app places no real trades.

A five-minute demonstration

Rehearse this flow once, with the application running on your laptop.

Before the jury: confirm FinBERT, inspect all pages, prepare the deck, and keep a reset replay available. The README documents browser visual checks that could not run in the build environment.

Time	Show	Explain
0:00-0:25	Overview	Required engine plus selected Module A.
0:25-1:05	Data sources	News and company feeds, fetch status and timestamps.
1:05-1:40	Live signal	Sentiment, event, impact and source evidence.
1:40-2:35	Synthetic replay	Fictional positive/negative events change weights.
2:35-3:15	Index allocation	10% baseline, current weights, total 100% and history.
3:15-4:20	Signal and policy	Confidence, severity cues, limits and active backend.
4:20-5:00	API and results	JSON export, implementation tests, evaluation limits.

If a feed fails

Show the Data sources error and move to the labeled replay. If live weights stay equal, explain that old/global/neutral/out-of-universe signals can be ineligible. Do not invent a market event or hide a fallback.

Useful text preview

Synthetic: Apple reports record profit and raises earnings guidance. Use Analyse text in the replay workspace. Preview only generates fields without saving. Save signal & rebalance records the text and updates the mock allocation.

The separate recorded-walkthrough script is in docs/VIDEO_WALKTHROUGH_10_MIN.md. Confirm its allowed length in the latest organizer document. This package does not contain a recorded video.

How the weights change

Measured allocation after twelve fictional events, with FinBERT active.

Stock	Equal baseline	After replay
AAPL	10.00%	12.50%
MSFT	10.00%	12.25%
NVDA	10.00%	12.43%
AMZN	10.00%	7.54%
GOOGL	10.00%	12.53%
META	10.00%	7.62%
JPM	10.00%	7.54%
XOM	10.00%	7.55%
JNJ	10.00%	12.53%
WMT	10.00%	7.51%
Total	100.00%	100.00%

These are mock allocations, not investment returns. Different replay times can create tiny recency differences.

Policy logic

Confidence, an assumed source coefficient, and time decay control each signal's strength. The target starts at equal weight and tilts with aggregated sentiment. It does not keep multiplying current weights with the same headline.

Defaults: 2% floor, 18% cap, 10% normal one-way turnover, 12-hour half-life, 72-hour maximum age, and minimum confidence 0.55. Neutral/global/ambiguous company signals do not drive company allocations. New mandatory bounds can require a compliance correction exceeding normal turnover.

Complete the submission

The files are prepared. Add your real public links and record your own demonstration.

- 1. Review the rules:** verify the full organizer document, exact deadline/timezone, required README headings, AI rules, video length, and official team details.
- 2. Publish the source:** create a public GitHub repository and upload the riskpulse project contents. Exclude .venv, node_modules, runtime, .env and tokens. Replace YOUR_USERNAME in the README clone command.
- 3. Review the deck:** docs/presentation.pdf has seven slides. The PPTX is editable. Confirm your name/team attribution.
- 4. Record and upload:** follow the ten-minute walkthrough script, subject to the latest organizer limit. Use actual app responses and your own narration. Test the video link signed out.
- 5. Fill the answer box:** use docs/SUBMISSION_TEMPLATE.txt, replace every placeholder with a real repository/video/deck link, and add the requested attachment.
- 6. Finish the portal:** Save, continue to the submission page, and complete final submission. Confirm submission success, rather than only a saved draft.

What to write in the answer box

The answer box is for your project summary and deliverable links. Briefly describe the risk engine, two source types, Module A, dashboard and API. Then include your public repository, accessible video, and presentation URL. The supplied template already contains the summary. Its link fields require your published URLs.

Results you can explain honestly

28 tests passed and a live check processed **50 items**. The 24 synthetic sanity cases matched sentiment/event labels. This is a small synthetic check, not a real-news accuracy or investment-performance benchmark. Mean loaded-CPU analysis time was 80.47 ms for those cases.

Codex assisted with implementation and materials. AI_USAGE.md records this. Personally review and understand the work, and follow the organizer's actual academic-integrity rules. No repository, video or portal submission has been published by this package.